

Digital Divide

Systematic inequality in access to, use of, and benefit from digital technology

GLOSSARY 05 - SHEET 8 / 8

PROFESSIONAL DEFINITION

The digital divide is the systematic inequality between individuals, groups, regions or states in access to digital technology, in the skills and autonomy to use it effectively, and in the outcomes obtained from that use. The literature distinguishes three levels: the first-level divide in physical access, the second-level divide in skills and usage, and the third-level divide in the tangible benefits realised (van Dijk, 2005; Scheerder, van Deursen and van Dijk, 2017).

WHO DEFINED IT ORIGINALLY

The term entered policy use through the United States National Telecommunications and Information Administration's Falling Through the Net reports from 1995. Jan A.G.M. van Dijk (2005) at the University of Twente developed the resources-and-appropriation theory; Paul DiMaggio and Eszter Hargittai (2001) introduced the second-level divide, and Alexander van Deursen and colleagues the third.

WHY THIS DEFINITION WAS DEVELOPED

Policy assumed that inequality would resolve as connectivity spread, which access statistics eventually confirmed while outcome inequality persisted or widened. Distinguishing three levels explained why: closing the access gap shifts the inequality to skills, and closing the skills gap shifts it to the benefits obtained, so the divide is displaced rather than eliminated.

FIGURE 1 - THREE LEVELS OF DIVIDE

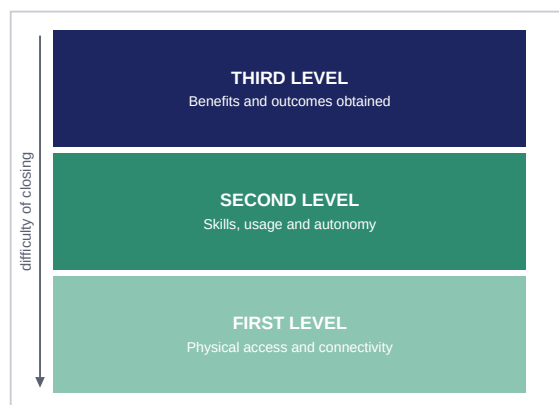


Figure 1. Closing a lower level displaces the inequality upward rather than removing it. This is why access-based measures show convergence while outcome-based measures do not, and why AI capability can widen inequality under conditions of universal access (after van Dijk, 2005).

HOW IT IS USED

Assess all three levels rather than access alone when appraising an AI-enabled service; universal access with unequal skills produces unequal outcomes and can widen rather than narrow the underlying inequality.

WHEN IT IS USED

In public-sector AI service design, in impact assessment of digital-only channels, in policy analysis of AI diffusion, and when a service is moved from a human to an automated channel.

BY WHOM IT IS USED

Public-sector service designers, policy analysts, corporate social responsibility and inclusion functions, and executives deciding to withdraw non-digital channels.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor framing was binary - connected or not - which made the problem appear soluble by infrastructure investment alone. The three-level model reconceives the divide as a sequence of displacements, each requiring a different intervention, and locates the durable inequality in outcomes rather than access. Applied to AI this predicts that a general-access technology can widen inequality if the capability to use it well is unequally distributed.

RELATED DEFINITIONS IN THIS COURSE

AI Literacy - Unit 2

Skill-Biased Technological Change - Unit 5

Externality - Unit 5

Algorithmic Bias - Unit 6

General Purpose Technology - Unit 5

SOURCES (HARVARD REFERENCING STYLE)

DiMaggio, P. and Hargittai, E. (2001) From the 'Digital Divide' to 'Digital Inequality'. Working Paper 15. Princeton, NJ: Center for Arts and Cultural Policy Studies.

Scheerder, A., van Deursen, A. and van Dijk, J. (2017) 'Determinants of internet skills, uses and outcomes: a systematic review of the second- and third-level digital divide', *Telematics and Informatics*, 34(8), pp. 1607-1624.

van Dijk, J.A.G.M. (2005) *The Deepening Divide: Inequality in the Information Society*. Thousand Oaks, CA: Sage.